

(after reset), followed by output O1 after the first clock pulse, then output O2 goes high after the second pulse, and so on, all the way up to O9. After the tenth pulse, it starts again from O0 and the cycle repeats. However, in this circuit, use of reset function at pin 15 forces the chip to repeat the cycle after the fourth pulse, instead of the tenth pulse. This is done by connecting output O4 (pin 10) to reset pin. Each output (O0 through O4) of IC2 goes high sequentially, and the cycle repeats. Around 12V is available across each output pin (O0 through O4), whenever it goes high. Output pulses turn-on transistors T1 through T4 sequentially, via base-current-limiting resistors R5 through R8.

An LED with a current-limiting resistor is connected across each relay coil. This indicates the working of the circuit. A pair of each relay contact is used to switch the video signal, while the other pair is used to switch the audio signal coming from the camera.

Switch S1 is used as a pause function. The pause facility is provided to stop the counting in IC2, so that you can hold and view a particular channel. During the pause action, we simply break the circuit by disconnecting pin 13 from the ground and allow resistor R3 to pull it high. The pause continues as long as you keep the button pressed. As soon as you release the switch, pin 13 is connected to ground again, which resumes the counting, and the video and audio sequencing continues.

Construction and testing

An actual-size, single-side PCB of the four-channel video and audio sequencer is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable plastic box.

Switch S1, potentiometer VR1 and

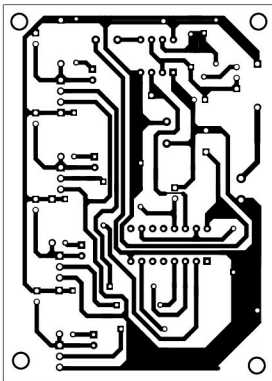


Fig. 2: An actual-size PCB pattern of the circuit

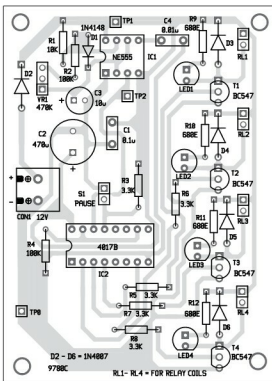


Fig. 3: Component layout of the PCB

the relays should not be mounted on the PCB. Connect the relays to respective connecting points (RL1 through RL4) provided in the PCB. The connection diagram for interfacing the cameras with relays and the TV monitor is shown in Fig. 4.

Author's prototype is shown in Fig. 5.

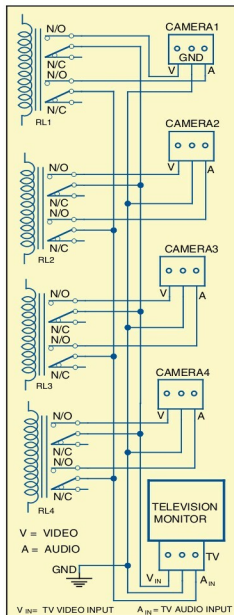


Fig. 4: Relay-connection diagram

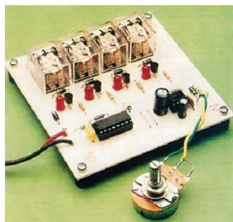


Fig. 5: Author's prototype

For troubleshooting, verify the parameters at various test points as shown in the table. ●

The author is an electronics hobbyist